Fiscal Year:	FY 2007	**Task Last Updated:**	FY 07/10/2007
PI Name:	Morris, Jeffrey F. Ph.D.		
Project Title:	Droplet formation processes from particle-laden liquids		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	10031	**Congressional District:**	4
Comments:	Professor F. Joseph Schork has acted as PI at Georgia Tech, while Dr. Jeff Morris (who was at Halliburton) remained technically responsible for the project. The grant is being transferred to CCNY where Dr Morris is joining the faculty January 1, 2005.		
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Fluid Physics 01-OBPR-02: Research and Flight Experiment Opportunities
Start Date:	04/01/2002	**End Date:**	01/31/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Nahra, Henry K	**Contact Phone:**	216-433-5385
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Flight Program:			
Flight Assignment:	NOTE: End date renegotiated to 1/31/2007, per PI (3/2007)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNC05GA13G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The study addresses basic drop formation processes for the dripping (or pendant drop) mode and laminar jet instability for solids-laden liquids, processes previously intensively studied for pure liquids. This work is building from that knowledge base by experimentally examining drop formation processes of noncolloidal suspensions at particle volume fractions up to, to date, 40% and eventually 50%. The tools used include high-speed video imaging and image analysis, with analytical tools from suspension mechanics and stability of fluid flow applied to the interpretation of the behavior. The primary issues are to determine the role of particle fraction and particle size, relative to the orifice diameter, in the drop formation process. The study is designed to isolate bulk rheological effects from finite particle size effects, in support of the goals of establishing the similarities and differences of the drop formation process and the associated fluid mechanical instabilities in a suspension of solid particles in relation to the equivalent processes in pure liquids. Numerical analysis using singularity methods to describe the particle effects on-axis in the jet are proposed for study. The basic physical processes in drop formation from mixtures are unexplored, while the phenomena are found in a number of applications. Thus the significance includes fundamental development of suspension mechanics in a free-surface flow with known instabilities, and also will provide guidance to the development of technologies for solid application through drop location processes. The most prevalent of these are inkjet processes in the electronics and textile industries, where the goal is to apply solids in fine lines for circuitry and imaging, respectively. Particle-laden "inks" raise issues in both cases because the very rapid pinch of small drops leads to a situation where the influences of rheology and isolated particle effects are largely unknown. Increased knowledge thus promises to open the way to new scientific approaches and at the same time the elucidation of basic processes should rapidly aid in the design of inkjet applications.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The benefits of the research toward technologies on earth can be expected in coating and inkjet type of applications. The basic science of mixture flows with free surfaces is generally poor, and is especially so when there is a rupture event such as drop formation involved: the presence of the particles renders the process significantly irregular and thus requires not only a reexamination using suspension mechanical concepts, but also statistical concepts. The advances made here will support understanding of inkjet printing with, for example, metal-laden liquids or ceramic particles, technologies under examination (largely by empirical approaches) for a range of applications.
Task Progress:	This report describes progress in the final year of the project. This project is now closed. The proposed research involved experiments, simulation and continuum modeling of the drop formation process. In the prior work, experimental study of drop forming systems in dripping and slow jetting for millimeter scale fluid columns and O(100) µm diameter particles was performed (Furbank & Morris 2004); fitting of the results to a two-stage continuum model (Furbank & Morris 2007) was shown to provide a reasonable description of the pendant drop thread thinning and rupture dynamics. Experimental study in the final year focused on considering smaller systems to determine the degree to which results at larger scale agree with these. As it proves quite difficult to match all dimensionless parameters across the scales, scoping of conditions outside the range of the large-particle suspensions, particularly in jetting, was performed. These results are still under analysis. The role of volume fraction of solids, phi, was shown to be delayed in its influence on the jet breakup relative to larger particle suspensions, a result we expect to be able to show is due to smaller capillary numbers at the smaller scales, leading to weaker surface deformation by particles. However, because surface tension drives capillary breakup of a pure liquid jet, there is apparently a competing set of effects which must be examined carefully before this conclusion is made. The bulk of effort in the final year of the project was directed toward numerical analysis using a semi-analytical technique we developed to account for the influence of discrete particles within a stretching column of viscous liquid. In this approach, implemented by post-doctoral research fellow Dr. M. Hameed (now of the University of South Carolina Upstate), the fluid motion in the jet was described by Stokes equations, with the particles described as force dipoles (also termed stresslets). The work only considers the particles to be on the axis of the jet. The surface boundary condition for a range of capillary numbers (Ca = µGB/sigma where µ is the fluid viscosity and sigma is the air-liquid surface tension, G is the stretching rate, and B is the initial radius of the jet) was imposed on the combined velocity field driven by the jet thinning and embedded force dipole, with the flow at the point of the dipole used self-consistently to determine its strength. The results provide what appears to be a good model of the evolution of the jet surface, with the influence of the particles on jet shape quite in agreement with observation. The ejection of isolated particles from the thread into the growing bulge regions is a feature seen in experiment and also predicted numerically. The results of this work are in preparation for the SIAM J. Appl. Math. A high school outreach effort was conducted by hosting a student from Stuyvestant High School of Brooklyn to the PI laboratory for a 3 month period. Mr. Daniel Bruno worked with Dr. Muhammad Hameed and the PI to design and perform experiments on the highspeed visualization of small (approximately 100 micron diameter) jets containing 5-10 micron diameter particles. These jets issued from a small-gauge needle with flow driven by syringe pump. The results of this particular work contribute to defining the particle loading at which behavior distinguishable from pure liquid is found. This work is part of Mr. Bruno's course on scientific research at Stuyvestant High and also forms the basis for his entry this year (2007) into the Intel Science Talent Search.
Bibliography Type:	Description: (Last Updated: 06/25/2007)
Articles in Peer-reviewed Journals	Furbank RJ, Morris JF. "Pendant drop thread dynamics of particle-laden liquids. " International Journal of Multiphase Flow 2007 Apr;33(4):448-68. http://dx.doi.org/10.1016/j.ijmultiphaseflow.2006.02.021 , Apr-2007
Articles in Peer-reviewed Journals	Dong H, Carr WW, Morris JF. "An experimental study of drop-on-demand drop formation." Physics of Fluids 2006 Jul;18(7):072102. http://dx.doi.org/10.1063/1.2217929 , Jul-2006
Articles in Peer-reviewed Journals	Dong H, Carr WW, Morris JF. "Visualization of drop-on-demand inkjet: drop formation and deposition. " Review of Scientific Instruments 2006 Aug; 77(8):085101. http://dx.doi.org/10.1063/1.2234853 , Aug-2006